The Cows Were Counted First: London's Great Smog and the Price of a Warm Room

A 1952 London fog killed about 12,000 people, but the first recorded casualties were prize cattle. A lecture on coal, politics, and the price of a warm room.

By Omer Okutan. ECON 238, University of Rochester. Modeled on Arnold Kling's "AIDS and a Blue Dress."

Two Documents

In the National Archives at Kew, in a box of old Ministry of Health papers, there is a short report written by a government veterinary scientist. Its title could not be plainer: The Effect of the Fog on Cattle at the Smithfield Show.

The Smithfield Show was the great annual exhibition of British livestock. In December of 1952 it was held at Earls Court in West London, and farmers had brought their finest animals to the capital to be judged: Aberdeen Angus and Hereford cattle, fat pigs, champion sheep, beasts that represented years of careful breeding and a great deal of money. In the second week of December the animals began to struggle for breath. Their tongues swelled. They wheezed in their pens. Around a hundred and sixty of them needed a vet, and about a dozen had to be slaughtered. Contemporary accounts say their keepers tried to protect them by holding sacking soaked in whisky over their noses, a homemade gas mask for a prize steer.

Those cattle were among the first casualties of the Great Smog of London to be written up, examined, and reported in the press. They were counted within days.

Now hold a second document beside the first. It is a death certificate, completed in December of 2020, at the coroner's court in Southwark, South London. It belongs to a nine-year-old girl named Ella Kissi-Debrah. Ella lived about twenty-five metres from the South Circular Road in Lewisham, one of the busiest roads in the city. In the three years before she died, she was taken to hospital almost thirty times with asthma attacks. She died in February of 2013. Seven years later, after her mother had fought to have an earlier inquest overturned, a coroner listed her medical causes of death. Acute respiratory failure. Severe asthma. And a third line: air pollution exposure. She is believed to be the first person in Britain to have air pollution written on a death certificate.

Sixty-eight years separate those two documents. In between, London's air killed people on a scale that is hard to take in. In those five days of December 1952, and the weeks after, the best modern estimate is that around twelve thousand Londoners died who would otherwise have lived. Not one of them, as far as anyone knows, had the air named as the cause of death. The cattle got a report in days. The people got a rough count weeks later, an argument that lasted decades, and a careful estimate only in 2001.

That is the puzzle at the heart of this story. Why did a great modern city find it easier to see a sick cow than a dead grandmother? And how did the same city, which went on to pass one of the most celebrated environmental laws in history and banish the killer fogs for good, arrive half a century later at a little girl killed by air that nobody could see at all?

The answer is a story about coal and about home. It is a story about a country that was broke, about a minister who was busy, and about a brewer with a passion for counting things. Underneath all of it is a single economic idea: pollution only becomes politics when it becomes visible, and when somebody is made to count it. For most of London's history, smoke was something you could see but not count. Today, much of our pollution is something we can count but cannot see. The Great Smog sits at the hinge between those two worlds, and to understand it we have to start long before the fog came down, in front of a fireplace.

A Civilisation Founded on Coal

In 1952, almost every home in London was heated the same way: an open coal fire in a grate, one in the sitting room and often more. The coal fire was not just a heating technology. It was the emotional centre of the British home. The family gathered around it. The kettle sat beside it. Children made toast at it on long forks. When Britons pictured comfort, safety, and belonging, they pictured a fire in a grate. The word "hearth" had a weight that "radiator" never would.

And behind every grate stood an industry that was close to the centre of national identity. In 1952 Britain mined around two hundred and twenty-eight million tonnes of coal. The coal industry had been nationalised in 1947, and the miners were heroes of the postwar settlement, the men who had fuelled the war and were now fuelling the recovery. Coal ran the power stations that lit London. It drove the steam trains that linked the capital to the country. George Orwell had put it bluntly fifteen years earlier, in The Road to Wigan Pier: "Our civilisation is founded on coal." He meant it literally.

London had been choking on that civilisation for centuries. Medieval Londoners complained about the smoke of sea coal, so called because it arrived by ship from the north. In 1661 the diarist John Evelyn presented King Charles the Second with a pamphlet called Fumifugium, a furious complaint about the coal smoke that blackened the city and, he argued, shortened its people's lives. Christopher Wren built St Paul's Cathedral in white stone, and it began turning grey almost at once.

By the nineteenth century, the fog had become part of London's personality. In 1853, Charles Dickens opened Bleak House with two words, "Fog everywhere," and then spent a page describing it creeping up the river, into the courts, into the lungs of old men by the fireside. Sherlock Holmes stalked through yellow fogs so thick that a man could vanish three steps away. Claude Monet came to London around the turn of the twentieth century to paint the Houses of Parliament dissolving in purple and orange haze, and he loved the fog, because it did astonishing things to light. Londoners gave their fogs nicknames: the London particular, the pea-souper, for its thick greenish-yellow

colour. Early in the twentieth century a London physician, Dr. H. A. Des Voeux, blended smoke and fog into a new word: smog.

Notice what the culture was doing. It turned a pollution problem into atmosphere, in both senses of the word. The fog was romantic, mysterious, quintessentially London. It was the price you paid for living in the greatest city on earth, and it seemed as natural as the rain.

It was not harmless, and people had evidence of that. A fog in December of 1873 pushed London's death rate forty per cent above normal. There were deadly fogs again in 1880, 1882, 1891, and 1892, and one in November of 1948. Economic historians have since tried to measure what Victorian coal smoke cost. Brian Beach and Walker Hanlon, using where factories were located and which way the wind blew, estimate that industrial coal smoke accounts for roughly a third of the extra mortality that people suffered for living in British cities in the 1850s. Hanlon has also found that cities that burned more coal grew more slowly over the following half-century, because smoke made them worse places to live and work. The smoke was a tax, paid in lungs and in lost growth. But nobody received a bill.

This is where the economics begins. When you light a coal fire in your sitting room, you get all of the benefit: the warmth is yours. The cost, the smoke, goes up your chimney and is spread across a million neighbours, a little bit each. Your fire doesn't kill anyone you could name. It just adds a fraction to an enormous shared cloud. Economists call this an externality, a cost that falls on people outside the decision. The air over London was a commons, a resource that belonged to everybody and therefore to nobody, and nobody had a reason to look after it. Even a Londoner who cared deeply about clean air gained almost nothing by giving up his own fire, because his chimney was one among millions.

Parliament had acted against smoke before, but only against one kind. The Smoke Nuisance Abatement Act of 1853 went after factory furnaces in London. Domestic chimneys were left alone. You could regulate a mill-owner. You could not tell an Englishman that the fire in his own home was a public nuisance. The hearth was private, and that boundary between the private fire and the public air would turn out to be the whole fight.

Then came the war, and the bill for it. Britain emerged from 1945 victorious and almost bankrupt, deep in debt and desperate for foreign currency. One of the few things it could sell abroad was coal, and good, hard, clean-burning coal fetched the best prices. So the best coal went abroad. At home, decent house coal was still rationed. What households could buy more freely was something called nutty slack: a cheap mix of coal dust and small broken lumps, high in sulphur and ash. It was not rationed, and the government encouraged people to burn it.

Sit with what that means. The pricing and rationing system of austerity Britain made the dirtiest fuel the cheapest and easiest to buy, and steered it straight into millions of London grates. This was not a conspiracy. It was a reasonable answer to a real problem, the country's urgent need to earn

money abroad, and it pushed the hidden cost onto the one thing nobody was pricing, which was the air.

One more change had just arrived. In July of 1952, London ran its last electric tram. Over three years the city had replaced its south London tram network with a modern fleet of diesel buses, at a cost of ten million pounds. Modernisation, in other words, had just added a new source of exhaust to the streets.

So picture London in late November of 1952. More than eight million people. Bomb sites still scattered through the East End. A population that had lived through the Blitz and was not easily frightened. Millions of grates burning cheap, sulphurous coal, coal-fired power stations inside the city itself, steam trains, factories, and a new fleet of diesel buses. Everyone believed, quite reasonably, that fog was weather, an act of God and an old London habit. All the city needed was the right weather.

Five Days in December

It arrived on Thursday, the fourth of December, 1952. A high-pressure system settled over southern England. The wind died. The weeks before had been bitterly cold, with heavy snow, and the cold now produced what meteorologists call a temperature inversion. Normally, air near the ground is warmer than the air above it, so it rises and carries smoke away. In an inversion, a layer of warmer air sits on top of cold air near the ground, like a lid on a pot. Nothing rises. Whatever the city breathed out, it would have to breathe back in.

And because it was cold, London did the natural thing. It built up its fires. More coal went into more grates, and more smoke went up more chimneys, and all of it stopped under the lid.

By Friday the fifth, the fog had turned from grey to yellow to something close to black. Over the next five days, visibility in parts of the city fell to a few feet. People said they could not see their own feet. Buses crawled through the streets behind conductors walking ahead with flashlights. Ambulances could not find addresses, so people walked to hospital. The only transport that ran normally was the Underground, because it was below ground and out of the fog.

The smog came indoors, too. On Saturday, the sixth, at Sadler's Wells theatre, a performance of Verdi's La Traviata was abandoned at the interval, because the audience in the back could no longer see the stage. At the White City stadium, greyhound racing was halted when the dogs could no longer see the mechanical hare they were supposed to be chasing. Some of the most vivid details of the week have that strange, almost comic quality, an opera cancelled for poor visibility, a race called off because the dogs were lost.

And there is one detail that tells you a great deal about class and geography in a smoky city. Up on Hampstead Heath, on the high ground in the north of London, people stood in sunshine and looked down on the smog lying across the city below them like a sea. Height was a kind of wealth. The

poor, the old, and the sick in the low-lying, crowded districts of the East End and along the river breathed the worst of it.

What were they breathing? Soot, first, millions of tiny carbon particles from the fires. But the real killer was the sulphur. Burning sulphurous coal produces sulphur dioxide, and in the damp, stagnant air it turned into a fine mist of dilute sulphuric acid, carried on soot particles deep into people's lungs. In 2016, a team of atmospheric chemists studying the haze that chokes Chinese cities today went back to the London smog and argued that nitrogen dioxide, also from burning coal, helped speed up the production of that acid inside the fog droplets. Scientists were still working out exactly how London's air killed its people more than sixty years after it did so.

And it did kill them. Deaths from bronchitis and pneumonia rose more than sevenfold. Death rates climbed in step with the levels of smoke, almost day for day. Most of the dead were elderly, or very young, or already suffering from heart and lung disease. They died at home, in their beds, by their fires, and in hospital wards across a city of eight million.

On Tuesday, the ninth, the wind returned from the west and swept the whole poisonous mass out over the North Sea. The sky cleared, and then something remarkable happened, or rather, did not happen. There was no great outcry. The newspapers had covered the fog mainly as a story about disruption: traffic, crime, cancelled football matches. The first estimate of its effect on health came only several weeks later.

Think about why. A bomb, a fire, a train crash, these produce bodies in one place, at one moment, with one obvious cause. The smog produced deaths that were spread across thousands of homes and dozens of hospitals, among people who were already old or ill and whose deaths looked, one at a time, like ordinary winter deaths. A doctor signing a certificate wrote bronchitis or heart failure or pneumonia, which was true. The smog was visible in the street and invisible on the paperwork. It could only be seen statistically, by adding up thousands of deaths and comparing them with what would normally have been expected. Statistically detectable, individually invisible: that is the signature of pollution.

Now return to the cattle at Earls Court. Why were they counted first? Because they were in one hall, at one show, attended by vets whose whole job was to examine them. Because they were valuable, and because they were property. Every prize steer had an owner who had lost money, and so its suffering was somebody's loss, recorded and reported. The lungs of Londoners belonged to no one in the same way. They were the human side of the commons. That is the first lesson of this story: we count what is owned.

The counting of the people, when it came, was a fight in itself. The Ministry of Health's official report attributed about four thousand extra deaths to the fog. But deaths in London stayed far above normal right through January and February of 1953. A preliminary government report, which was never finalised, suggested that those later deaths were due to an influenza epidemic, a convenient

explanation, since flu is nobody's fault. Nearly fifty years later, in 2001, the epidemiologists Michelle Bell and Devra Davis went back to the records, to death registers, hospital admissions, and even health insurance claims, and rejected the flu explanation. They estimated that about twelve thousand people died as a result of the smog between December 1952 and February 1953. A follow-up study in 2004 found that, to explain those later deaths with influenza, you would need an epidemic roughly twice as deadly and four times as widespread as anything doctors were actually seeing that winter.

So the most famous environmental disaster in British history was officially undercounted by about two thirds, and the full count took half a century. Hold that thought, because the government that had to decide what to do about the smog had nothing like that count. It had a fog that had come and gone, a few thousand dead whose death certificates said nothing about the air, and a very long list of other priorities.

The Minister, the Brewer, and the Moustache

In December of 1952, Winston Churchill was Prime Minister again, leading a Conservative government that had come to power the year before. Responsibility for the smog fell largely on two ministers. And if you want to understand why nothing happened quickly, you have to understand what those men were trying to do, because they were not fools and they were not villains.

The Minister of Housing and Local Government was Harold Macmillan. His great task was houses. Britain had lost hundreds of thousands of homes to German bombs, and millions of families lived in overcrowded, crumbling slums. The Conservatives had promised to build three hundred thousand new homes a year, and Macmillan staked his career on hitting that number. He did hit it, in 1953, and it made him a political star. Every pound, every ton of steel, every hour of a minister's attention went to that target.

Now look at the smog from Macmillan's desk. Serious action would mean telling millions of families, in the middle of austerity, that they had to rip out their fireplaces and buy new heaters and more expensive fuel. It would mean burning the good coal at home instead of selling it abroad for the currency the country badly needed. It would mean a fight with an industry that employed hundreds of thousands of men. And the benefit was cleaner air, which was hard to measure, would arrive slowly, and would not show up in any statistic a minister could announce. The costs were concentrated, visible, and immediate. The benefits were diffuse, invisible, and delayed. That is the classic shape of a problem that democratic governments postpone.

So Macmillan postponed it. He argued in public that the problem was complex and expensive and that broad economic considerations had to be weighed. In private, he is reported to have grumbled that people now expected the government to solve every problem, and that "smog" had somehow captured the imagination of the press and public. The Minister of Health, Iain Macleod, was just as

impatient. He was reported to have complained that anyone would think fog had only started in London since he became Minister.

It is worth pausing on Macleod, because he embodies how ordinary smoke was in that world. A year or so later, Macleod held a press conference to announce the government's acceptance of the new research linking cigarettes to lung cancer, and he reportedly chain-smoked through it. This was a society that lived in smoke, in its pubs, its offices, its homes, and its streets. Smoke was simply what the air was like.

The government's most visible response to the disaster was a mask. In late 1953 the Ministry of Health arranged for doctors to be able to prescribe gauze smog masks on the National Health Service to patients with heart and lung conditions. The masks did almost nothing against sulphur dioxide gas. But they were cheap, and they looked like action. The Spectator magazine, a year after the smog, observed that the only defence London had been given was the little gauze mask, and estimated that the smog might have cost the city as much as two million pounds a day in damage and delay. When the real remedy is expensive and the political reward is small, governments are tempted to buy the appearance of action instead. A mask is that kind of purchase.

The second classic move was to appoint a committee. In July of 1953, under growing pressure in Parliament and the press, the government set up a Committee on Air Pollution, and here the story takes its turn. To chair it, they chose a man named Sir Hugh Beaver.

Beaver was a civil engineer by training, a practical man who had spent his career building things, and he had become managing director of the Guinness brewery. He was not an environmentalist, a crusader, or a scientist of the air. He was a businessman who liked facts. And in a sense he was the ideal person to answer the question the government had dodged, because he set about counting the cost. His committee's report, published in November of 1954, did not treat smog as weather or as the price of greatness. It treated it as a preventable waste. Smoke was unburned fuel, money literally going up the chimney. It corroded buildings, ruined laundry, darkened the city, and sickened its people, and the committee put the economic damage at something on the order of hundreds of millions of pounds a year. Most importantly, Beaver's committee crossed the line Parliament had refused to cross for a century. It said domestic chimneys had to be included. The private fire was a public matter.

There is a lovely coincidence here. In the very same years, Beaver was involved in another project. According to the story Guinness itself tells, he had got into an argument during a shooting party about which was the fastest game bird in Europe, and found that no reference book could settle it. So he commissioned one. The first Guinness Book of Records was published in August of 1955, less than a year after his smog report. The man who wanted a book that would settle arguments with counted facts was the same man who settled the argument about London's air by counting its costs. A committee that was meant to bury the problem became the instrument that forced it into the open.

Even then, the government dragged its feet. What finally moved it was a backbencher. Gerald Nabarro was a Conservative member of Parliament from Kidderminster, a flamboyant self-made businessman famous for his enormous handlebar moustache and his love of publicity. He introduced a private member's bill on clean air, the kind of bill that ordinary MPs can propose but rarely pass. It embarrassed the government into bringing forward its own legislation. The Clean Air Act received royal assent on the fifth of July, 1956.

Look at the cast of characters who delivered Britain's great environmental law. Not a green movement; there wasn't one yet. A Conservative government that had resisted. A brewer who liked facts. A showman with a moustache. And Harold Macmillan, the minister who had played down the problem, became Prime Minister just six months after the Act passed, and his government was the one that had to carry it out. History rarely sorts its people into heroes and villains, and this one doesn't. The people who delayed had real priorities, and the people who acted were not the ones you would have expected.

What did the Act actually do? It banned dark smoke from chimneys. It required new furnaces to be made smokeless. And, most importantly, it allowed local councils to declare "smoke control areas", districts where it became an offence to burn anything but approved smokeless fuels such as coke, gas, or electricity. Inside those areas, householders could get grants covering seventy per cent of the cost of converting their fireplaces, with central government paying the larger share. In other words, the law did not just forbid. It also paid people to change. That is a crucial detail, because it means Britain solved its commons problem with a combination of rules and money: everyone in a district had to switch at once, so no one bore the cost of being the only clean chimney on the street, and the state helped pay for the switch.

What Actually Cleared the Air

Here is the version of the story you usually hear: the smog killed thousands, the country was shocked, Parliament passed the Clean Air Act, and the killer fogs ended. It is a satisfying story. It is also too simple, and the ways it is too simple are the most useful part.

First, the fogs did not stop. A smog in January of 1956, before the Act had even passed, killed around a thousand people. Another in December of 1957 killed several hundred more. And in December of 1962, ten years almost to the week after the Great Smog, London disappeared again, and somewhere around seven hundred people died. The smoke that week was only about a third of what it had been in 1952, but the sulphur dioxide was nearly as bad.

Why so slow? Because the Act left the hard part to local councils, and councils had to decide, street by street, to declare smoke control areas. Many dragged their feet. Smokeless fuel was in short supply. Conversion costs rose. In the industrial north especially, people resisted. A public health inspector who helped create a large smoke control area in Leeds later remembered that persuading residents was a miserable job. Their faces fell when he told them they would no longer

be allowed to burn coal, and brightened only when he got to the part about the grant. The British hearth did not go quietly. In 1968, Parliament had to pass a second Clean Air Act, partly to push the councils that had done almost nothing.

Second, and more surprising: much of what cleared London's air was not the law at all. It was prosperity. In 1957, the new Prime Minister, Harold Macmillan, the same man who had dismissed the smog, told the British people that most of them had "never had it so good," and in material terms he was right. Wages rose through the late 1950s and 1960s. Families who had once been grateful for a coal fire began to want something cleaner and easier, a gas fire you could switch on, an electric heater, eventually central heating. Nobody enjoyed hauling coal up from the cellar and cleaning out ash every morning.

There is a wonderful piece of evidence for this in the reports of London's local medical officers of health. In Hammersmith, officials surveyed homes in the earliest smoke control areas, where the council had paid to install new grates designed to burn coke, the approved smokeless fuel. They found that many households had since ripped out those subsidised grates, at their own expense, to put in gas, electric, or oil heating. The government had paid people to make a small change. People, with more money in their pockets, chose to make a bigger one.

Then technology moved. Gas production changed, and in the late 1960s Britain began to bring natural gas ashore from under the North Sea, cheap and clean and piped right into the home. The slums cleared, and new housing came with modern heating. Power stations moved out of the city centre. Coal's share of home heating collapsed.

So what is the honest verdict on the Clean Air Act? It did not single-handedly clean the air. It deserves its fame for something more important. It broke the old boundary. It established, in law, that the fire in your own home was a matter of public concern, and it gave every district a way to switch together, so that clean air no longer depended on millions of separate sacrifices. It worked best where it went with the grain of what people were starting to want anyway. Economists sometimes say that as societies get richer, they demand cleaner environments and can afford to pay for them, and Britain in the 1960s is a textbook example. But the delay was not free. Every year that councils hesitated, people died in fogs that a faster policy might have prevented. The lesson is not that markets would have fixed it on their own, or that law fixed it on its own. It is that law and rising incomes did it together, and that the law decided how many people would die while we waited.

One more twist deserves mention. Some of the cleanup simply moved the pollution. The new rules encouraged tall chimneys on power stations, which pushed sulphur high into the sky, where the wind carried it east. By the 1970s, Scandinavian scientists were tracing acid rain falling on their lakes and forests back to British smokestacks. The externality had not disappeared. It had crossed the North Sea.

The Fire Comes Home

Now come back to the present, and to the second document.

Ella Kissi-Debrah did not die in a fog. On the days she was rushed to hospital, the air in Lewisham may have looked perfectly clear. The pollution that the coroner found had contributed to her death was mainly nitrogen dioxide and fine particulate matter from traffic, and both are almost invisible. There is a painful irony in that, too. In the early 2000s, British governments pushed drivers toward diesel cars, because diesel engines burned less fuel and released less carbon dioxide, which helped with climate targets. Diesels also produced far more nitrogen dioxide and soot. Like the coal exports of 1952, it was a reasonable solution to one real problem that pushed the cost onto the air.

In 1952, pollution was easy to see and hard to count. Today, it is the other way around. We have monitoring stations, satellites, and computer models that can estimate pollution street by street. What we no longer have is the fog. Nobody misses a bus because of nitrogen dioxide. No opera is stopped at the interval. And so the counting, which in 1952 was the missing piece, has become the thing people argue about.

You can see that in London's recent politics. To cut traffic pollution, the city created an Ultra Low Emission Zone, which charges older, more polluting vehicles a daily fee to drive in it. In 2023 the Mayor of London expanded it to cover the whole of Greater London, out into the suburbs. The backlash was ferocious. Some people vandalised the zone's cameras. Online, the policy became tangled up with conspiracy theories about governments trying to control where people could travel. In a by-election in the outer London seat of Uxbridge that summer, the Conservatives held on in a race widely interpreted as a vote against the zone's expansion. The costs, a daily charge on an old car that a tradesman relies on for work, were concentrated, visible, and immediate. The benefits, cleaner air and fewer asthma attacks spread across millions of people, were diffuse, invisible, and delayed. That is the same shape Harold Macmillan faced in 1953. The difference is that in 1953 nobody could argue the fog wasn't there. In an era of social media, polarised trust, and suspicion of experts, people can and do argue that the numbers aren't real.

And then there is the most unexpected return of all. In the last fifteen years, the wood-burning stove has become one of the most fashionable objects in a British home. It appears in magazine spreads of cosy cottages and smart London terraces, a symbol of warmth, authenticity, and a certain kind of comfortable, rural-flavoured affluence. Stove sales boomed. And by the government's own estimates for 2022, domestic burning, mostly of wood, produced around twenty-nine per cent of Britain's emissions of fine particulate matter, more than road transport. Those figures are disputed. The stove industry argues that the government overestimated how much wood people burn, and later official revisions put wood burning's share lower, closer to a sixth. But even at the lower figure, it is a striking fact. The hearth, the very thing the Clean Air Act took on, has come back, not as the fuel of the poor, but as a luxury for the comfortable. The externality is exactly the same as it was in 1952: the warmth is yours; the smoke is your

neighbour's.

So return, at the end, to the two documents.

The cattle at Smithfield were counted within days because they had owners, and their suffering was somebody's loss. The twelve thousand people who died in the Great Smog took half a century to count, because the air they breathed belonged to no one, and their deaths, one by one, looked like ordinary winter. Ella Kissi-Debrah was counted, by name, on a death certificate, because her mother refused to accept the first inquest, fought to have it overturned, and forced the state to look. In each case, the air became visible only when someone was made to count its cost: a vet, a brewer, two epidemiologists, a grieving mother.

The Great Smog of 1952 did something terrible and something useful. It killed on a scale that could not be ignored, and in doing so it made the cost of a warm room visible to a whole nation. It forced Britain to accept that a private fire is a public matter, and that idea, more than any single clause of the law, is its legacy. But the fog was also a gift that our pollution no longer gives us. The poison of 1952 announced itself. The poison of today is quiet. What we can see, we eventually regulate. What we can count, we argue about. And what we can neither see nor count, we simply pass on to the next generation, as Londoners did for seven hundred years, until the air turned black and the cows began to cough.

Reading List

• The Great Smog of 1952, UK National Archives blog: https://blog.nationalarchives.gov.uk/the-great-smog-of-1952/

• The Great Smog of 1952, Met Office case study: https://weather.metoffice.gov.uk/learn-about/weather/case-studies/great-smog

• Michelle Bell and Devra Davis, Reassessment of the lethal London fog of 1952, Environmental Health Perspectives (2001): https://pmc.ncbi.nlm.nih.gov/articles/PMC1240556/

• Bell, Davis and Fletcher, A retrospective assessment of mortality from the London smog episode of 1952 (2004): https://pmc.ncbi.nlm.nih.gov/articles/PMC1241789

• Brian Beach and W. Walker Hanlon, Coal Smoke and Mortality in an Early Industrial Economy, Economic Journal (2018): https://ideas.repec.org/a/wly/econjl/v128y2018i615p2652-2675.html

• W. Walker Hanlon, Coal Smoke and the Costs of the Industrial Revolution, NBER: https://www.nber.org/papers/w22921

• Stephen Mosley, Clearing the air: can the 1956 Clean Air Act inform new legislation?, History & Policy: https://historyandpolicy.org/?p=1616

• The Great London Smog of 1952: its consequences and contemporary relevance, University of London Press: https://read.uolpress.co.uk/read/law-and-justice-in-the-1950s/section/b6a4942b-4d97-4622-91c0-bcfea82e016a

• Hammersmith Medical Officer of Health report, Wellcome Collection: https://wellcomelibrary.org/moh/report/b18238300/43

• Clean Air Act 1956: https://en.wikipedia.org/wiki/Clean_Air_Act_1956

• Sir Hugh Beaver: https://en.wikipedia.org/wiki/Hugh_Beaver

• Coroner rules air pollution played part in Ella Kissi-Debrah death (PA, 2020): https://guernseypress.com/news/uk-news/2020/12/16/coroner-rules-air-pollution-played-part-in-ella-kissi-debrah-death

• New research raises concern about domestic wood burning (2024): https://envirotecmagazine.com/2024/12/09/new-research-raises-concern-about-domestic-wood-burning-the-uks-biggest-source-of-pm2-5/

• Arnold Kling, AIDS and a Blue Dress: https://arnoldkling.substack.com/p/aids-and-a-blue-dress